

Homework 1

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- introduce county, basically manhattan.
- people of color in certain neighborhoods, also overlaps w low income
- introduce diesel particulate matter - pg 16 on pdf
- describe health effects
- analysis: if u compare the two maps the red parts are like in teh same parts of the county
- propose some explanations for why there is so much overlap
- list equity issues with said overlap
- restrictions: diesel pm was actually calculated at the census tract level, not block

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(sf)
```

Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE

```
library(here)
```

here() starts at /Users/eriny/Documents/GitHub/eds223/EDS223-HW1

```
library(tmap)
```

Registered S3 method overwritten by 'stars':

```
method          from  
st_interpolate_sf sf
```

```
# read in geodatabase of EJScreen data at the Census Block Group level
```

```
ejscreen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI"))
```

Reading layer `EJSCREEN\_StatePctiles\_with\_AS\_CNMI\_GU\_VI` from data source

```
`/Users/eriny/Documents/GitHub/eds223/EDS223-HW1/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI`  
using driver `OpenFileGDB`
```

Simple feature collection with 243021 features and 223 fields

Geometry type: MULTIPOLYGON

Dimension: XY

Bounding box: xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350

Projected CRS: WGS 84 / Pseudo-Mercator

```
# filter to a state you are interested in
```

```
state <- ejscreen %>%
```

```
  dplyr::filter(ST_ABBREV == "NY")
```

```
# filter to a county you are interested in
```

```
county <- ejscreen %>%
```

```
  dplyr::filter(CNTY_NAME %in% c("New York County"))
```

```
# find the average values for all variables within counties
```

```
#state_by_county <- aggregate(state, by = list(county$CNTY_NAME), FUN = mean)
```

```
tm_shape(county) +
```

```
  tm_polygons(
```

```
    fill = 'PEOPCOLORPCT',
```

```
    fill.scale = tm_scale(values = c('green', 'red')),
```

```
    fill.legend = tm_legend(title = 'Percentage People of Color'),
```

```
    fill_alpha = 0.6
```

```
  ) +
```

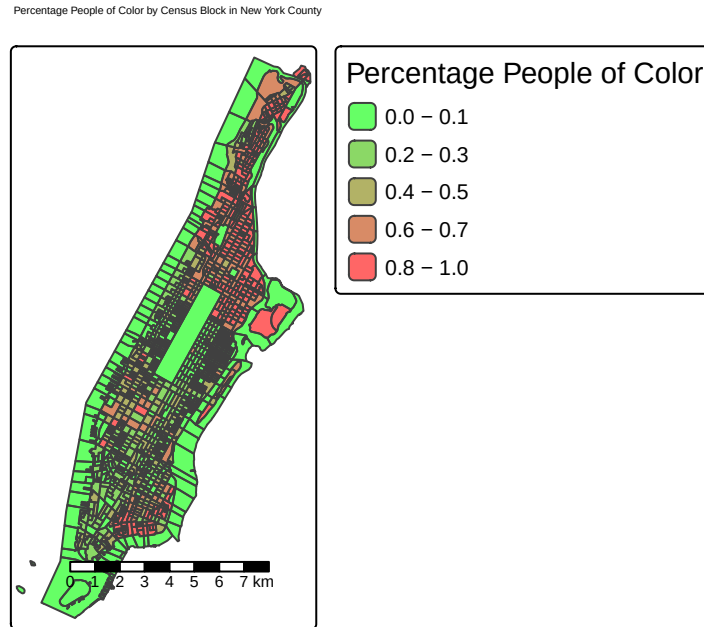
```
  tm_scalebar() +
```

```
  tm_title('Percentage People of Color by Census Block in New York County', size = 0.5)
```

Multiple palettes called "green" found: "kovesi.green", "tableau.green". The first one, "kovesi.green", is used.

[plot mode] fit legend/component: Some legend items or map components do not fit well, and are therefore rescaled.

i Set the tmap option `component.autoscale = FALSE` to disable rescaling.



```
tm_shape(county) +
  tm_polygons(
    fill = 'P_DSLPM',
    fill.scale = tm_scale(values = c('green', 'red')),
    fill.legend = tm_legend(title = 'Percentile Diesel Particulate Matter (g/m3)'),
    fill_alpha = 0.6
  ) +
  tm_scalebar() +
  tm_title('Percentile Diesel Particulate Matter by Census Block in New York County', size
```

Multiple palettes called "green" found: "kovesi.green", "tableau.green". The first one, "kov

[plot mode] fit legend/component: Some legend items or map components do not fit well, and are therefore rescaled.

i Set the tmap option `component.autoscale = FALSE` to disable rescaling.

